SHRI RAMSWAROOP MEMORIAL UNIVERSITY

PROBLEM SET – IV

Session	: 2025-26(Even Sem.)	**Semester**	: VI
Course	: BCA & B.Sc IT	**Subject Code**	:UCS6124
Branch	: Normal & Specialization	**Subject Name**	: Machine Learning Elective-II
Group	: ALL	**Name of Faculty Member**	: Dr. Archana Sachindeo Maurya, Er. Aakriti Nigam, Er. Prem Kumar Yadav
Unit	: IV	**Topic Covered**	: Unsupervised Learning

S. No.	Questions
1	What is unsupervised learning? How does it differ from supervised learning? Discuss its advantages and limitations.
2	How does the clustering process work? Explain it in detail and describe different clustering techniques with examples.
3	Explain the K-means algorithm with a step-by-step procedure and illustrate it using a small numerical example.
4	Why is K-means clustering widely used? Discuss its advantages and disadvantages, and mention situations where it fails.
5	Define the Expectation-Maximization (EM) algorithm and explain its steps and role in clustering.
6	Compare K-means and EM algorithms in terms of working principle, assumptions, and applications.
7	Describe Gaussian Mixture Models (GMM) and explain how they differ from K-means clustering.
8	Illustrate how the EM algorithm is applied to GMM, including mathematical intuition and iterative steps.
9	Analyze the role of probability in GMM-based clustering and differentiate between soft and hard clustering.
10	Discuss the limitations and challenges of clustering techniques, especially in high-dimensional data and real-world scenarios.

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PROBLEM SET – V

Session	: 2025-26(Even Sem.)	**Semester**	: VI
Course	: BCA & B.Sc IT	**Subject Code**	:UCS6124
Branch	: Normal & Specialization	**Subject Name**	: Machine Learning Elective-II
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Unit	: V	**Topic Covered**	: Graphical Models

S. No.	Questions
1	What are Bayesian Networks? Explain their structure and components with an example.
2	How does conditional independence work in graphical models? Illustrate with a suitable diagram.
3	Why are Markov Random Fields important in probabilistic modeling? Discuss their applications.
4	Explain the concept of exact inference in graphical models. What methods are commonly used?
5	Differentiate between Bayesian Networks and Markov Random Fields with respect to structure and inference.
6	Describe sequential graphical models and their significance in real-world problems.
7	Compare the Max-Sum and Max-Product algorithms in terms of working and applications.
8	Illustrate the working of Hidden Markov Models (HMM) with a suitable example.
9	Analyze the role of Conditional Random Fields (CRF) in sequence modeling tasks.
10	Discuss the challenges involved in performing inference in large graphical models.

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